

Class 3: XG Architecture Basics

RAN, Core, and O-RAN

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CIS 4930/5930 Future Edge Networks
xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks

What does the icon actually tell you?



Ericsson active 5G antennas

Turn off Wi-Fi for ten seconds.

What appears: LTE, 5G, 5G+, or 5G UW?

- **LTE:** 4G radio access.
- **5G:** 5G radio is available.
- **5G+:** AT&T mid- or high-band.
- **5G UW:** Verizon C-band or mmWave.

LTE = Long Term Evolution mmWave = millimeter wave

Q: Does that icon describe the radio, the core, or the whole network?

A: Only the radio access, and even that is filtered through carrier branding. It says nothing about the core behind it, and nothing about the speed you actually get.

Somebody paid about \$50 for that icon this month

What a line costs

- Unlimited starts at \$45–\$60
- Same price at 2 GB or 60 GB
- Bundled video and hotspot

Entry unlimited plans, autopay, August 2026.

What is not on the bill

- A price per gigabyte
- A speed you are owed
- Which radio you get

Q: If your bill does not change when your usage doubles, what are you buying?

A: Access, not data. You are buying the right to attach at all, which is why carriers compete on bundles instead of on megabits.

Now multiply that by a nationwide build

Money coming in

Revenue per phone line is flat, and in places falling.

AT&T postpaid phone ARPU was about \$56.64, down 0.8% in a year.

AT&T Q3 2025; Verizon 2025 guidance.

ARPU = Average Revenue Per User capex = capital expenditure

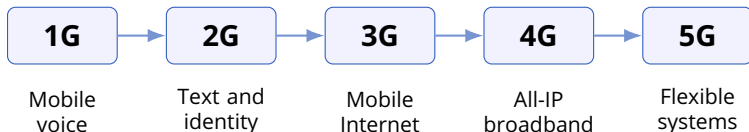
Money going out

Verizon guided 2025 capital spending to \$17.5–18.5 billion.

That is up from \$17.1 billion the year before.

Flat revenue per user, rising spend. How long does that work?

Each generation made a new habit feel normal



The app was visible.

The architecture underneath was not.

The first mobile call was basically a flex



Martin Cooper with a 1973 DynaTAC prototype

Martin Cooper called his rival.

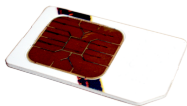
- Cooper was a Motorola engineer.
- In 1973 he called Joel Engel.
- Engel led AT&T's cellular team.
- AT&T was betting on car phones.
- Motorola bet on a handheld.
- Service reached users in the 1980s.

1G moved the telephone away from the wall.

The SIM card is the part that knows you



Nokia 3310



GSM SIM card

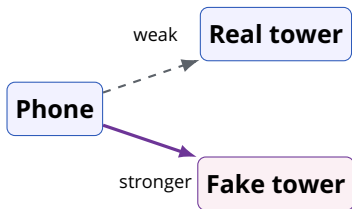
2G went digital.

- Digital voice and SMS
- Identity moved onto the SIM
- Mass-market roaming

SMS = Short Message Service SIM = Subscriber Identity Module

Q: Which one is “you” to the network: the phone, the SIM, or the number?

A: The SIM. It stores the subscriber identity and the secret key used to authenticate you. The number is only a label the network maps onto that identity.



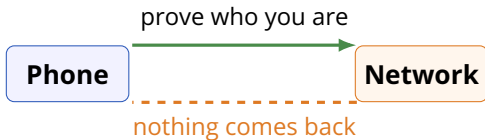
Three steps.

- The phone attaches to the strongest signal.
- The tower asks who it is.
- The phone answers with its identity.

IMSI = International Mobile Subscriber Identity

Equipment built to do this has a name: an IMSI catcher.

2G checked your ID. You never got to check theirs.



Q: The SIM proves the phone is real. What proved the tower was real?

A: In 2G, nothing. Authentication ran one direction only. Later generations add mutual authentication, but the phone still has to pick a tower before it can check anything. We trace what each generation did about this on Wednesday.

3G brought video calls before eye contact



Nokia 6680 commercial (YouTube)

Data became central.

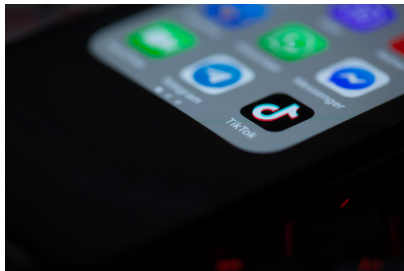
- Web, email, multimedia
- UMTS/WCDMA radio access
- Packet data grew beside voice

UMTS = Universal Mobile Telecommunications System

WCDMA = Wideband Code Division Multiple Access

3G made the phone a data device.
The core still ran voice and data as two separate worlds.

4G made buffering feel like a personal insult



TikTok app on phone

Broadband became the default.

- Streaming and short video
- Maps, ride sharing, live apps
- Voice moved to IP with VoLTE

VoLTE = Voice over LTE IP = Internet Protocol

4G did not invent video. It made video normal.

The 5G icon can lie by omission



Trying AT&T's "5G E" (YouTube)

- It names a radio, not a speed.
- Branding sits on top of it.
- Spectrum and congestion decide.
- A 5G radio may use a 4G core.

Q: “Why is LTE faster than 5G here?”

A: A congested 5G cell on a narrow band can lose to a clean LTE cell. AT&T's old "5G E" icon was LTE-Advanced wearing a 5G name.

Paraphrased user question, 2026

You press Call. What happens next?



UMTS video call

Your phone has to find a tower.
The network has to recognize you.
Then it has to find the other end.

Q: Which part of the network handles each of those three jobs?

A: The RAN finds the tower and holds the radio link. The core recognizes you and decides what you may do. The core also reaches the other end, through the Internet or the telephone network.

Four worlds are hiding behind one tap



RAN

Turns radio waves
into managed links.

Core

Remembers who you are,
where you are,
and what you may do.

UE = User Equipment (your phone) RAN = Radio Access Network
PSTN = Public Switched Telephone Network, the traditional telephone network
3GPP = 3rd Generation Partnership Project, the body that writes these standards

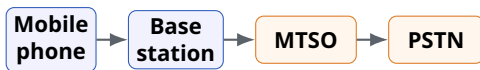
Conceptual model based on 3GPP system architecture

The network still has the same six jobs

- 1 Find a cell
- 2 Request access
- 3 Prove identity
- 4 Get radio resources
- 5 Route voice or data
- 6 Move without dropping

The node names change.
The jobs are surprisingly persistent.

1G connected radio voice to the telephone network



A circuit stays reserved while the call is active.

MTSO = Mobile Telephone Switching Office

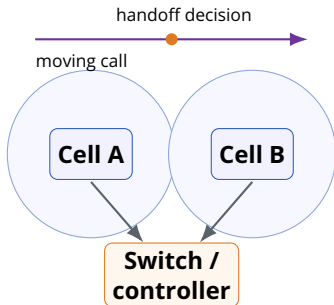
PSTN = Public Switched Telephone Network

The path was a circuit.

- Analog radio in the cell
- MTSO linked the towers
- PSTN carried the call

A mobile call was still a telephone call.

Mobility means changing the network while using it



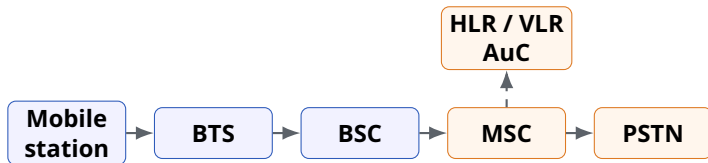
A moving car raises three questions:

- Who measures the signal?
- Who picks the next cell?
- Who keeps the call alive?

Q: What goes wrong if the handoff is late? If it is too early?

A: Late, the call drops while the old cell fades. Too early, the phone ping-pongs between cells and spends signaling on a move it did not need.

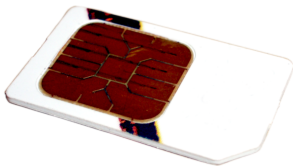
2G split the base station into radio and control



BTS Base Transceiver Station
BSC Base Station Controller
MSC Mobile Switching Center
HLR / VLR Home / Visitor Location Register
AuC Authentication Center

Digital identities made roaming scalable.

The SIM became a network passport



GSM SIM card

Home network

knows your subscription, number, and credentials.

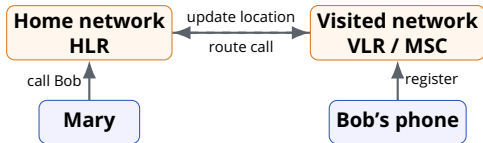
Visited network

knows where to reach you right now.

Q: If you replace the phone but keep the SIM, what changed in the network?

A: Almost nothing. The network still authenticates the same subscriber record, so your number, your subscription, and your roaming state survive. Only the device identity and its radio capabilities changed.

A call can find you even when you leave home



Bob visits another city.

- 1 Bob registers there.
- 2 The HLR is updated.
- 3 Mary's call is routed.

Mobility also needs a database.

GPRS was an Internet sidecar

GPRS = General Packet Radio Service, packet data added onto GSM.

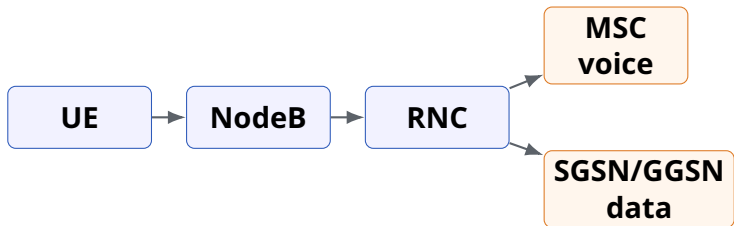


SGSN = Serving GPRS Support Node GGSN = Gateway GPRS Support Node

- Voice still used the MSC path.
- Packet data used SGSN and GGSN.
- The phone gained a standing route to the Internet.

Packet data arrived next to voice, not instead of it.

3G changed the radio faster than the architecture



NodeB = 3G radio node • RNC = Radio Network Controller

- UMTS/WCDMA raised mobile-data capability.
- NodeB and RNC echoed the BTS/BSC split.
- Circuit voice and packet data lived side by side.

[3GPP 3G overview](#)

Voice

Circuit switched

MSC and the telephone world

Data

Packet switched

SGSN/GGSN and the Internet

Running both worlds at once worked. It was not simple.

Can you match the tower name to the generation?

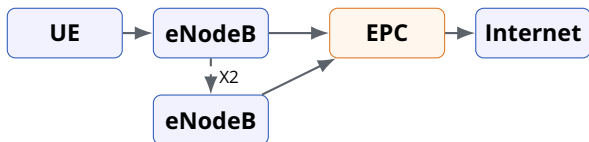
- | | |
|----------|-----------|
| ① BTS | • 2G GSM |
| ② NodeB | • 3G UMTS |
| ③ eNodeB | • 4G LTE |
| ④ gNodeB | • 5G NR |

GSM = Global System for Mobile Communications NR = New Radio
The prefixes are generational: e = evolved (LTE), g = next generation (5G).

Q: Which generation removed the separate radio controller?

A: 4G. The eNodeB absorbed the scheduling and handover control that the RNC used to own, so LTE has no separate controller box in the main path.

4G flattened the RAN

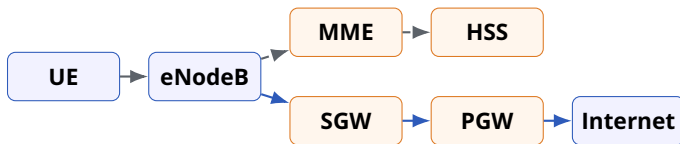


No separate RNC sits in the main LTE path. EPC = Evolved Packet Core

- The eNodeB absorbed the radio control that lived in the RNC.
- Neighboring eNodeBs coordinate directly over X2.
- The core became the Evolved Packet Core.

Fewer boxes in the path.
More logic inside the base station.

The EPC turned the core into packet plumbing



dashed = control signaling • solid blue = user data

MME Mobility Management Entity

HSS Home Subscriber Server

SGW Serving Gateway

PGW Packet Data Network Gateway

Learn the jobs, not the acronym list.

Wait — where did SGSN and GGSN go?

What it did

SGSN tracked where you were and kept your data session alive. GGSN was the door out to the Internet.

Each box did both jobs at once: it made decisions **and** it carried your traffic.

What broke

Traffic grew far faster than decisions did.

Ten times the video does not mean ten times the number of attach requests. But because one box did both, carrying more data meant buying more decision-making you did not need.

What replaced it

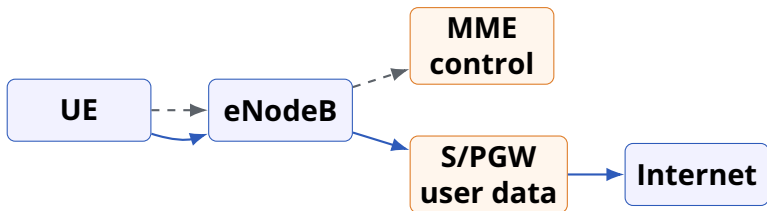
4G cut each box along that seam.

MME decides and never touches your packets. S-GW and P-GW move packets and decide nothing.

Now each side grows on its own.

The box did not fail. It was doing two jobs with one growth rate.

Control asks; the user plane moves packets



Control plane

attach, authenticate,
move, establish a bearer

User plane

carry the voice, video,
game, or web packets

Separating the two planes
lets each one scale on its own.

And the MSC? It had been there since 2G.

What it did

The MSC was a telephone switch that happened to serve mobiles.

It set up a circuit for each call and held that circuit open until somebody hung up.

What broke

LTE carried IP and nothing else.

A circuit switch has no place in a packet network. Keeping one meant running a second network beside the first, with its own equipment, its own staff, and its own bill.

What replaced it

Nothing, at first — and that was the problem.

Voice had to stop being a network service and become an application riding on IP, like any other app.

So who carries your voice call on an all-IP network?

In 4G, a phone call had to become an Internet app

Early LTE

Data on LTE

Voice falls back to 2G/3G

VoLTE = Voice over LTE IMS = IP Multimedia Subsystem, the part of the core that sets up and controls calls

VoLTE

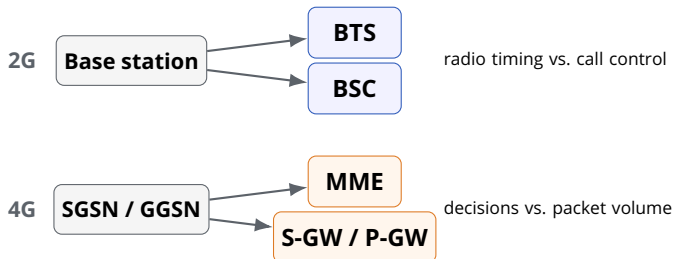
Voice as a managed IP service

IMS provides call control

Q: Why was “all-IP” a bigger change than making the radio faster?

A: Speed is one number. All-IP changed the structure: one packet transport carries everything, and voice becomes an application on top of it instead of its own circuit-switched network.

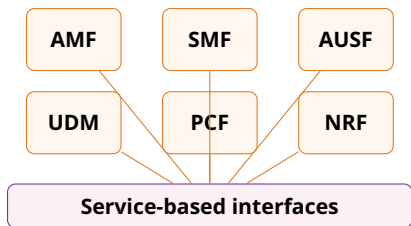
You have now watched the same move twice



One box becomes two when its halves need to change at different speeds.

On Wednesday you will watch it happen twice more — to the core, and then to the base station itself.

5G made the core look more like a cloud platform



AMF Access + Mobility
SMF Session Management
UPF User Plane Function
AUSF/UDM identity
PCF policy
NRF service registry



The SMF steers the UPF over the N4 interface.

[Nokia 5G Core Service Based Architecture](#)

Core functions now discover and call each other as services.

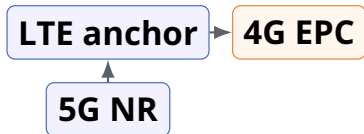
A 5G radio does not guarantee a 5G core

NSA = Non-Standalone

SA = Standalone

QoS = Quality of Service

NSA



The phone shows 5G.
Control still runs through
LTE and the 4G core.

SA

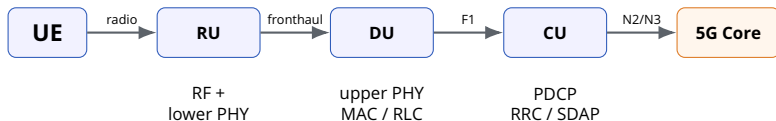


A 5G radio on a 5G core.
Slicing, edge, and new QoS
become possible.

The icon reports the radio.
The features come from the core.

AT&T: "Not all 5G is created equal," 2025

The 5G base station can be split across locations



One gNodeB can be a system spread across a tower, a cabinet, and a data center.

RU / DU / CU = Radio / Distributed / Central Unit

PHY = physical layer MAC = Medium Access Control RLC = Radio Link Control

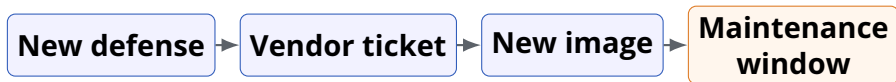
PDCP = Packet Data Convergence Protocol RRC = Radio Resource Control

SDAP = Service Data Adaptation Protocol

Q: Who owns each box, and who owns the interfaces between them?

A: In a classic RAN, one vendor owns all of it. Splitting the base station only helps if the seams between the pieces are specified well enough for a second vendor to plug in.

Can base-station security become an app?



You invent a defense against a new attack on the base station.

Q: Who installs it, who approves it, and what happens at 2:00 a.m.?

A: In a classic RAN the vendor answers all three. Your idea becomes a support ticket, then a firmware image, then a scheduled maintenance window — and nothing moves fast at 2:00 a.m.

O-RAN's bet:
some network behavior can ship as a tested application.

Four RAN models, four different questions



Virtualization

What runs where?

D-RAN = Distributed RAN C-RAN = Centralized (or Cloud) RAN
vRAN = Virtualized RAN O-RAN = Open RAN

Disaggregation

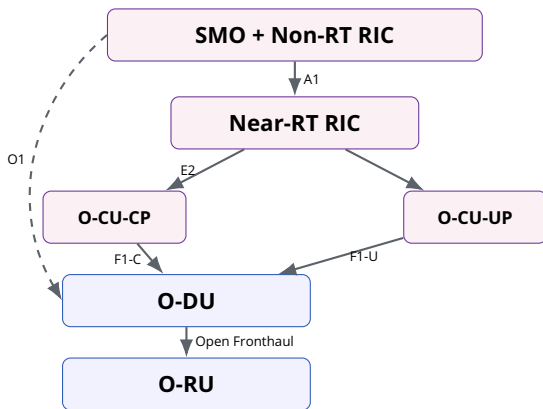
Which functions split?

Open interfaces

Who can connect?

Cloud software alone does not create an open,
multi-vendor RAN.

O-RAN opens the seams, then adds control loops



RAN chain

O-CU-CP / O-CU-UP
O-DU / O-RU

Control

SMO + Non-RT RIC
Near-RT RIC

Open seams

A1, E2, O1, Open
Fronthaul

SMO = Service Management
and Orchestration

RIC = RAN Intelligent
Controller

Non-RT / Near-RT =
non- and near-real-time

CP / UP = control / user plane

O-RAN ALLIANCE architecture, redrawn from 2023 white paper

Open RAN is not the same as open-source RAN

Open RAN

Published interfaces
Disaggregated functions
Multi-vendor is possible

Open source

Source code is available
License permits reuse
Interface may stay closed

Q: If two boxes claim the same interface, will they work when plugged together?

A: Not automatically. A specification leaves options, versions, and timing choices open. Conformance and interoperability testing — plugfests, lab integration — is what finishes the job.

[O-RAN ALLIANCE architecture and resources](#)

Opening the RAN creates both options and work

Options

- More vendor choice
- Faster experiments
- Programmable control

Work

- Fronthaul and timing
- Integration testing
- Bigger security surface

Q: Would you rather debug one black box or five open interfaces?

A: Both are work, in different places. One box hides the failure and hands the whole fix to one vendor. Five interfaces make the failure visible, and make the integration yours.

[O-RAN ALLIANCE overview \(2026\)](#)

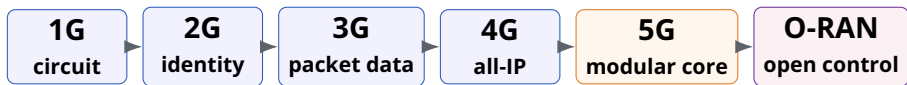
Five generations in one low-resolution map

	Radio	RAN control	Core	Mode	Example
1G	Base station	MTSO	Analog core	Circuit	Voice in a car
2G	BTS	BSC	MSC + HLR/VLR	Circuit; GPRS packet	SMS, roaming
3G	NodeB	RNC	CS + PS cores	Both	Mobile web / video
4G	eNodeB	In the eNodeB	EPC, all-IP	Packet	Maps, apps, streaming
5G	gNodeB	CU (RRC), split CU/DU/RU	Service-based 5GC	Packet + QoS flows	Flexible services + edge

CS = circuit-switched PS = packet-switched QoS = Quality of Service
EPC = Evolved Packet Core 5GC = 5G Core

The radio name changes.
The deeper story is where control lives.

Follow the same action one last time



The packet still needs **radio access**,
identity + policy + routing,
and now sometimes **programmable control**.

Each generation relocates functions.
O-RAN changes who may control them.

Exit ticket: draw the boundaries



- 1 Mark the **RAN** and **Core** boundary.
- 2 Add one **O-RAN** interface or controller.
- 3 Answer in one sentence:
What changes next—radio, core, or control?

Appendix: the acronym decoder ring

Era	Node	Meaning / job
1G	MTSO	Mobile Telephone Switching Office; switching and mobility control
2G	BTS / BSC	Base Transceiver Station / Base Station Controller
2G/3G	MSC, HLR, VLR, AuC	Voice switch; home/visitor state; authentication
GPRS	SGSN / GGSN	Packet mobility + gateway to external packet networks
3G	NodeB / RNC	Radio node / Radio Network Controller
4G	eNodeB	LTE radio node; absorbs most RNC control
4G	MME, HSS, S-GW, P-GW	Mobility; subscriber data; serving and packet gateways
5G	gNodeB	5G radio node, often split into CU / DU / RU
5GC	AMF, SMF, UPF	Access/mobility; session control; packet forwarding
5GC	AUSF, UDM, PCF, NRF	Authentication; subscriber data; policy; service registry
O-RAN	SMO / RIC	Management/orchestration and intelligent control

Generic terms:

3GPP = 3rd Generation Partnership Project IP = Internet Protocol

QoS = Quality of Service PDU = Protocol Data Unit

CS / PS = circuit- / packet-switched CP / UP = control / user plane

Appendix: interface map

Uu UE ↔ radio: air

S1 / NG RAN ↔ core

X2 / Xn neighbor handover

F1 O-CU ↔ O-DU

Open FH O-DU ↔ O-RU

E2 Near-RT RIC ↔ RAN

A1 Non-RT RIC ↔ Near-RT RIC

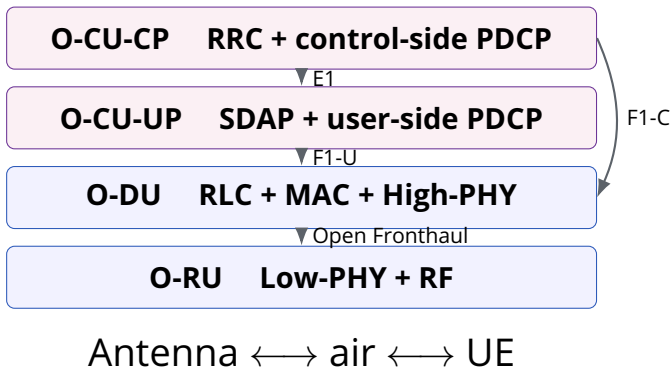
O1 SMO ↔ managed functions

O2 SMO ↔ cloud platform

Interfaces expose coordination;
they do not remove it.

[O-RAN ALLIANCE architecture and resources](#)

Appendix: where the protocol stack is split



A split is a placement decision.
Latency, bandwidth, compute, and ownership all move with it.

Appendix: handover changed owners, not purpose

Era	Main controller	What crosses the boundary?	What must survive?
1G	MTSO	Analog voice circuit	The conversation
2G	BSC / MSC	Radio channel + circuit state	Call and subscriber context
3G	RNC / core	Radio bearer; CS or PS state	Voice or packet session
4G	eNodeB + EPC	IP bearer and mobility context	All-IP session
5G	gNodeB + 5GC	PDU session, QoS flows, policy	Service continuity
O-RAN	Same 3GPP functions	RIC may influence parameters	Safe control under open APIs

Q: Which is harder to move—a radio channel, a packet tunnel, or its policy?

A: The channel has the hardest deadline; the policy is the hardest to keep consistent across cells, cores, and vendors.

Sources and further reading

- **3GPP TS 23.501**

Authoritative 5G system architecture.

- **Nokia: 5G Core service-based architecture**

Readable service-based-architecture role map.

- **O-RAN Architecture and Resources**

Interfaces, RICs, and current specifications.

- **O-RAN ALLIANCE overview (2026)**

Current official overview.

- **AT&T 5G Standalone announcement (2025)**

Why a 5G icon can appear before a 5G core.

- **Course references**

Local materials in `reference/` were used for background only, not reproduced as slide images.

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